Week 1

Fred Azizi

2022-08-31

Who am I?

My name: Farideddin Azizi (You can call me Fred)

My pronouns: He/him/his

I am a PhD student in the Department of Statistics.

What are we going to do here?

- Structure of the class
- Q/A- Working on worksheet in class- Working on additional worksheet in your time.
- Roughly, this is how the class time will be divided into:
  - ~10 to 20 min: Q/A about the material.
  - ~ 30 to 40 min: Work on the new worksheet.

Questions

Questions:

1- How is this class going to affect my grade?

2- Is attendance required?

3- How do I get the slides/worksheets?

4- Will you provide the solutions?

Where do I get slides/worksheets/solution

Please scan this QR code:

SCAN ME

Or simply visit:

fredazizi.github.io

Chapter 1: Quick Review (1)

"*Data* are just the raw facts and figures that pertain to a measurement of interest.

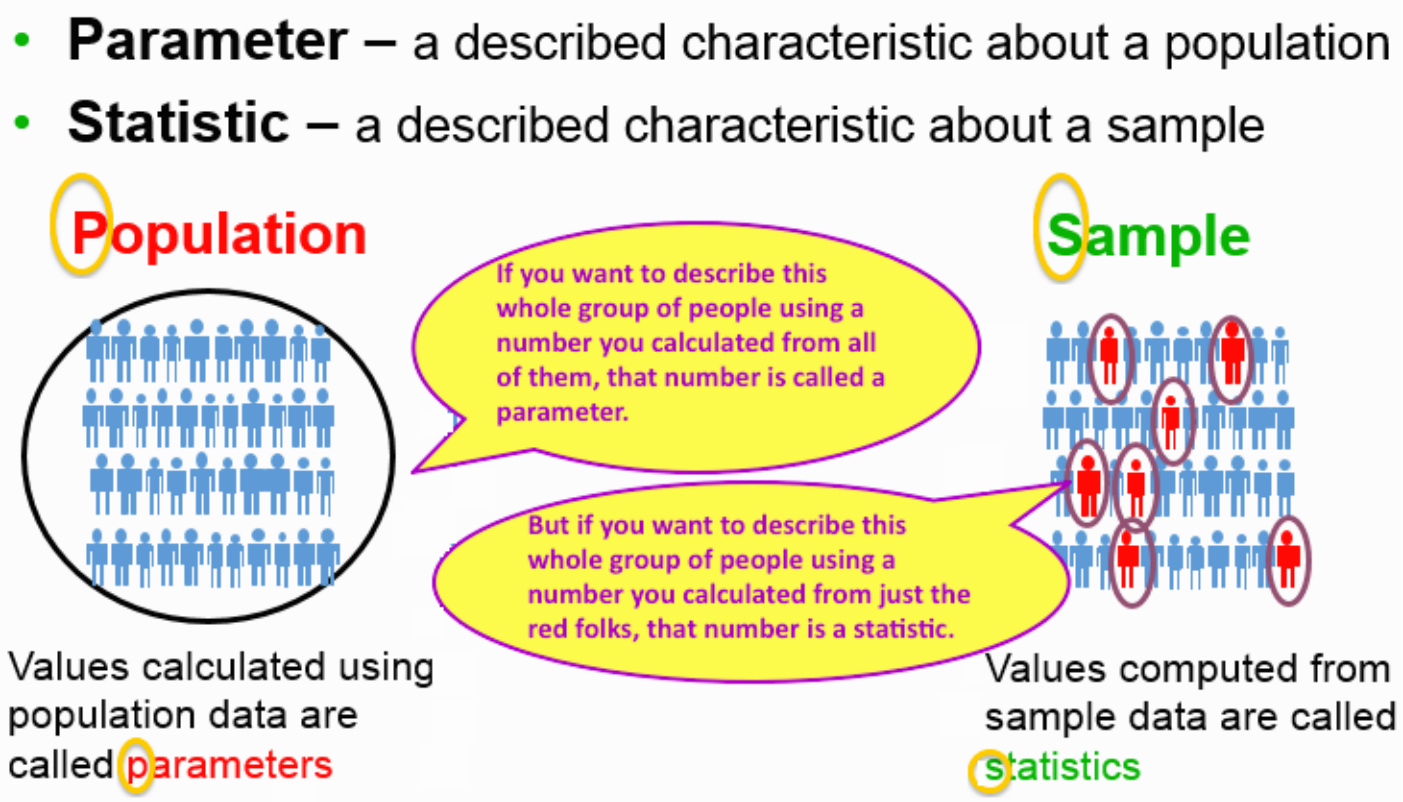
Information [...] is derived from the facts for the purpose of making decisions.

One of the main reasons to use statistics is to transform *data* into *information*."

Chapter 1: Quick Review (2)

Population: All possible subjects that are of interest to us in a particular study. (Note: not necessarily a group of people)

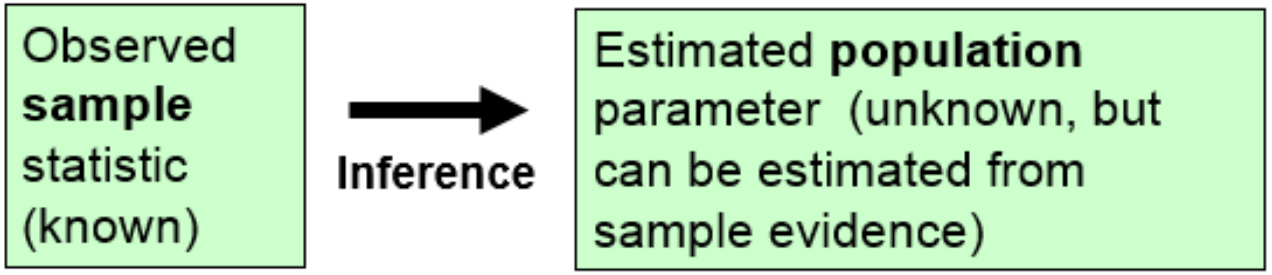
Sample: A portion of the population that is representative of the population from which it was selected.



Chapter 1: Quick Review (3)

Descriptive statistics describe/summarize data.

Inferential statistics make claims or draw conclusions about a population based on a sample.



Question: Using Smartphone Data to Monitor Health

Researchers at MIT gathered data from cell phones of student volunteers, such as location and phone use. They found they could reliably conclude when students were ill, as *"sick students moved around much less than normal, and depressed students had fewer calls and text messages."*

Is this an example of (A) Descriptive or (B) Inferential statistics?

Are the student volunteers the (A) Sample or (B) Population?

Are data such as average texts per day for this group an example of a (A) Statistic or (B) Parameter?

Worksheet

Lets take a look at the in-class worksheet.

Chapter 1: **key Concepts**

- Descriptive Statistics: Descriptive statistics deals with methods of organizing, summarizing, and presenting data in a convenient and informative way.
  - You can use numerical techniques to summarize data (mean, median, variance, etc)
  - You can use graphical techniques to present data (Bar chart, scatterplot, words cloud, etc)
- Inferencial Statistics: Inferential statistics is a body of methods used to draw conclusions or inferences about characteristics of populations based on sample data.
- A population is the group of all items of interest to a statistics practitioner.
- A descriptive measure of a population is called a parameter.
- A sample is a set of data drawn from the studied population.
- A descriptive measure of a sample is called a statistic.